3-Bit Counter Driven by a 1 Hz Divided Clock

Item	Detail
Target Board	ZedBoard (Xilinx Zynq-7000, xc7z020clg484-2)
Design Tool	Xilinx Vivado Design Suite
Source Clock	100 MHz on-board oscillator (pin Y9, GCLK)
PLL Output Clock	8 MHz (Clocking Wizard IP, clk_wiz_0)
Final Clock	1 Hz (custom clock-divider module)
Output	3-bit up-counter (counter[2:0]) driving LD0–LD2

1. Objective

The objective of this assignment is to design and implement a 3-bit binary counter on an FPGA whose transitions are slow enough to be visually observed on the board's LEDs. Since the board's native clock (100 MHz) is far too fast for direct visual observation, the design uses a Phase-Locked Loop (PLL) to generate an intermediate 8 MHz clock, followed by a digital clock-divider circuit that further reduces this to a 1 Hz clock. This 1 Hz clock drives the 3-bit counter, so each LED toggles at a rate a human eye can follow.

2. Design Overview

The system is composed of four functional blocks connected in a clock-generation and consumption chain:

- 100 MHz on-board oscillator connected to the FPGA global clock input— the board-level 100 MHz reference clock, connected to the FPGA's global clock pin.
- Clocking Wizard / PLL (clk_wiz_0) — a Xilinx IP core that locks onto the 100 MHz input and synthesizes a stable 8 MHz output clock (clk_out1), together with a 'locked' status signal.
- Clock Divider (DIV) — a free-running counter that divides the 8 MHz PLL output down to a 1 Hz square wave.
- 3-bit Counter (counter) — a simple up-counter clocked at 1 Hz, whose 3-bit output is wired directly to three on-board LEDs.

The top-level module (Top.v) instantiates and wires these four blocks together, and the 'locked' output of the PLL is used as the active-low-style reset qualifier for the downstream logic, ensuring the divider and counter only start counting once the PLL clock is stable.

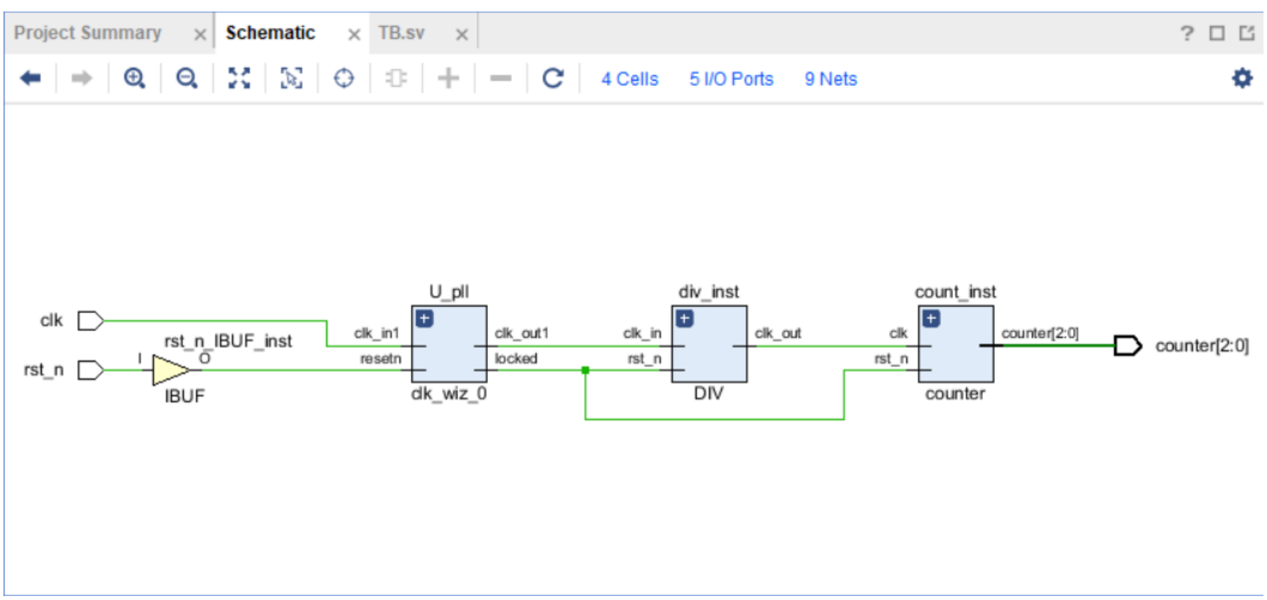

Figure 1. RTL / elaborated schematic showing the PLL (U_pll), clock divider (div_inst) and 3-bit counter (count_inst) chained together, generated in Vivado.

3. Module-Level Design

3.1 Top-Level Module — Top.v

The top module simply wires the PLL, divider and counter together. Reset (rst_n) is an external active-low push-button/switch input, and the PLL's 'locked' signal (rst_from_pll) is used as the synchronizing reset for the divider and counter, so both downstream blocks stay held in reset until the PLL clock is valid.

3.2 PLL — clk_wiz_0 (Xilinx Clocking Wizard IP)

The 8 MHz clock is generated using Vivado's Clocking Wizard IP (instantiated as clk_wiz_0). This is a vendor-supplied PLL/MMCM wrapper rather than hand-written RTL: it takes the 100 MHz board oscillator as clk_in1, and is configured to output an 8 MHz clock on clk_out1. It also exposes a locked output, asserted once the PLL has achieved phase lock, which the rest of the design uses as its reset-release condition.

3.3 Clock Divider — DIV.v

The divider is a free-running 23-bit binary counter clocked by the 8 MHz PLL output. Every time the counter reaches 3,999,999 it resets to zero and toggles the output register clk_out; otherwise it simply increments. Toggling the output every 4,000,000 input clock edges produces a 50%-duty-cycle square wave whose frequency is derived below.

3.4 3-bit Counter — counter.v

The final stage is a straightforward synchronous 3-bit up-counter, clocked by the 1 Hz divided clock and asynchronously reset by the same PLL-lock-derived reset used elsewhere in the design. It increments by one on every rising edge of the 1 Hz clock and wraps around naturally from 3'b111 back to 3'b000 due to the fixed 3-bit width.

With a 1 Hz clock, each bit of the counter toggles at a different, human-observable rate: counter[0] toggles every 1 s, counter[1] every 2 s, and counter[2] every 4 s — giving a full 8-second repeating pattern across the three LEDs.

5. Pin Constraints (cons.xdc)

The constraints file assigns the design ports to physical ZedBoard pins and sets the I/O standards for each bank. The key constraints for this design are:

Signal	Pin	Board Function	I/O Standard
clk	Y9	GCLK — 100 MHz on-board oscillator	LVCMOS33 (Bank 13)
rst_n	F22	SW0 — reset switch	LVCMOS18 (Bank 35)
counter[0]	T22	LD0	LVCMOS33 (Bank 33)
counter[1]	T21	LD1	LVCMOS33 (Bank 33)
counter[2]	U22	LD2	LVCMOS33 (Bank 33)

The XDC file also defines the primary clock and false-path constraints:

```
create_clock -name clk -period 10 [get_ports clk]   ; 100 MHz reference

set_false_path -from rst_n -to counter_reg[0]/CLR
set_false_path -from rst_n -to counter_reg[1]/CLR
set_false_path -from rst_n -to counter_reg[2]/CLR

set_false_path -from counter_reg[0]/C -to counter[0]
set_false_path -from counter_reg[1]/C -to counter[1]
set_false_path -from counter_reg[2]/C -to counter[2]
```

The false-path constraints correctly tell the timing engine not to analyze the asynchronous reset path or the slow (1 Hz) counter-to-output-pin path as a critical timing path, since both are inherently asynchronous / far below the FPGA's operating speed.

6. Implementation Results

6.1 Device (Floorplan) View

After synthesis, placement and routing in Vivado, the implemented design maps to the following region of the xc7z020clg484-2 device. Because the design is extremely small (one PLL, one 23-bit divider counter, one 3-bit counter), it occupies a very small fraction of the fabric, visible as the sparse green routing at the bottom-left of the device view.

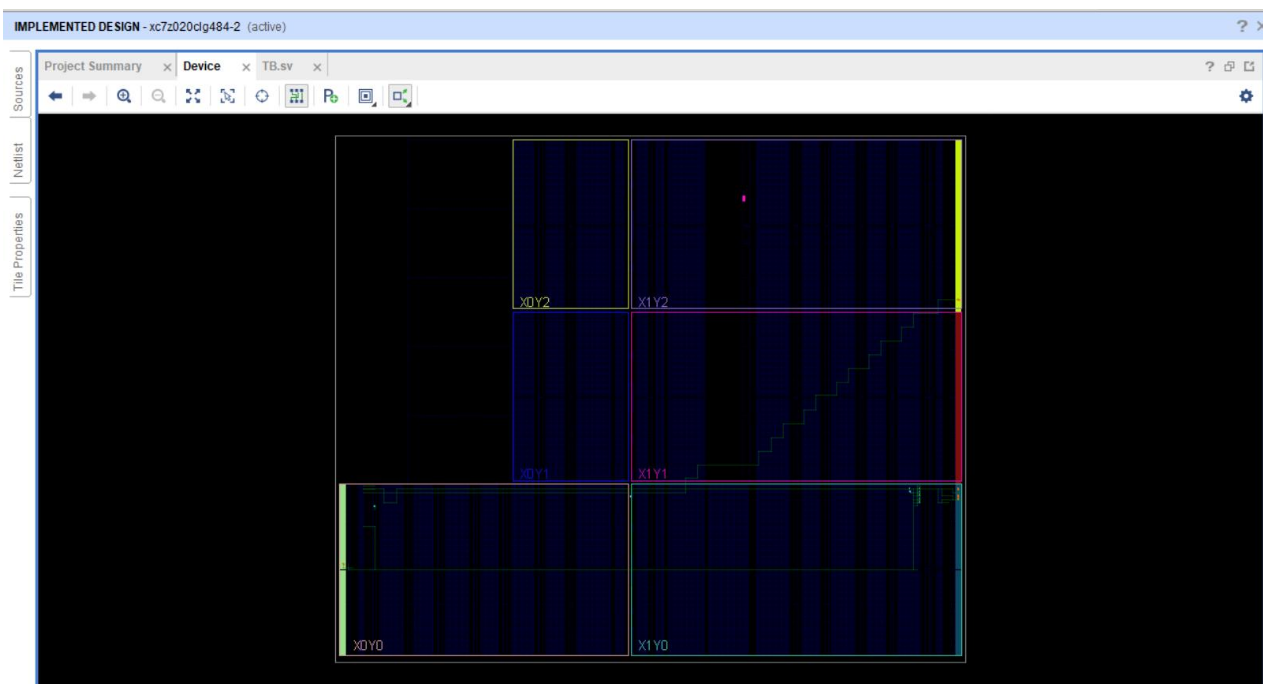

Figure 2. Post-implementation device (floorplan) view showing the placed and routed design on the Zynq-7000 fabric.

6.2 Resource Utilization and Power

Post-implementation utilization confirms the design consumes negligible fabric resources — well under 10% of every configurable logic resource — since it contains no user-clocked logic beyond three flip-flops and a 23-bit divider register. The one PLL/MMCM instance used by the Clocking Wizard is the dominant contributor to both logic utilization (25% of available PLL/MMCM blocks) and dynamic power (about 90% of the 0.109 W dynamic power budget), which is expected since PLLs are analog/mixed-signal blocks that draw a comparatively fixed amount of power regardless of the surrounding logic size.

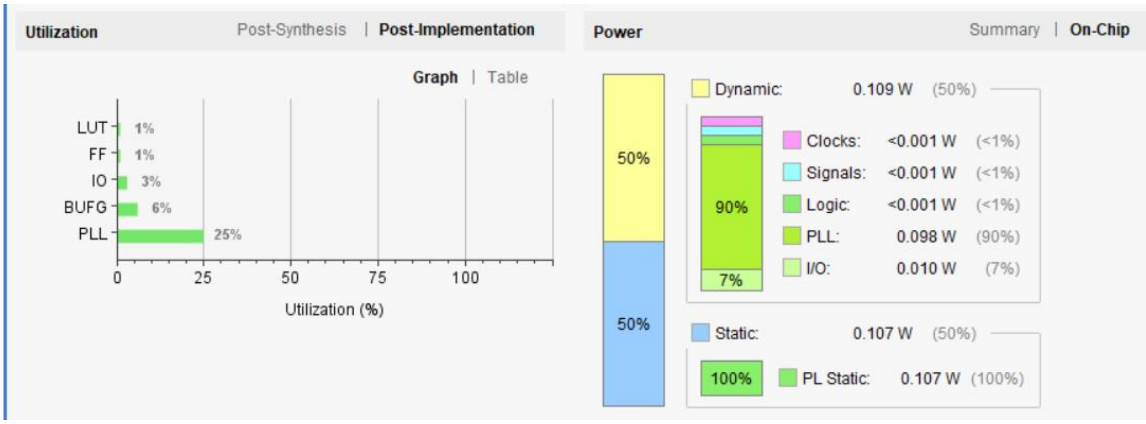

Figure 3. Post-implementation resource utilization (LUT, FF, IO, BUFG, PLL) and on-chip power breakdown.

Resource	Utilization
LUT	1%
FF (Flip-Flop)	1%
IO	3%
BUFG	6%
PLL	25%

The detailed utilization report below gives the exact resource counts behind the percentages above,

Resource	Utilization	Available	Utilization %
LUT	22	53200	0.04
FF	27	106400	0.03
IO	5	200	2.50
PLL	1	4	25.00

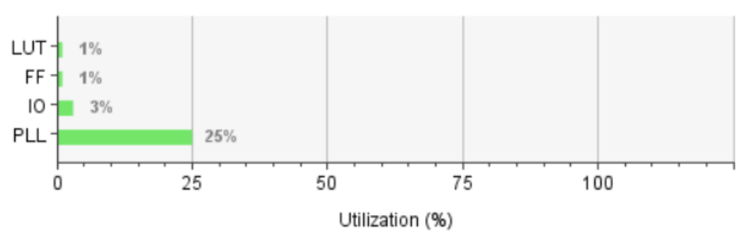

Figure 3a. Detailed post-implementation utilization report - resource counts, device totals and utilization percentage.

Resource	Utilization	Available	Utilization %
LUT	22	53,200	0.04%
FF	27	106,400	0.03%
IO	5	200	2.50%
PLL	1	4	25.00%

Power Metric	Value	Share
Total Dynamic Power	0.109 W	50%
Total Static Power	0.107 W	50%
Dynamic — PLL	0.098 W	90% of dynamic
Dynamic — I/O	0.010 W	7% of dynamic
Dynamic — Clocks / Signals / Logic	< 0.001 W each	< 1% each

6.3 Timing Analysis

Static timing analysis (STA) was run after implementation across all 24 timing endpoints in the design (the divider bits, counter bits and PLL-related paths). All three checks — setup, hold and pulse-width — pass with positive slack and zero failing endpoints, confirming the design meets timing at its constrained 100 MHz input clock with comfortable margin, which is unsurprising given how far the actual functional clock rates (8 MHz and 1 Hz) sit below the fabric's maximum speed.

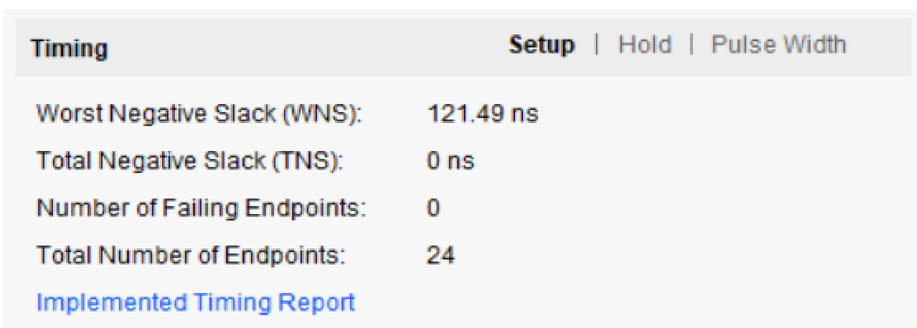

Figure 4. Setup timing summary — 121.49 ns worst negative slack, 0 failing endpoints.

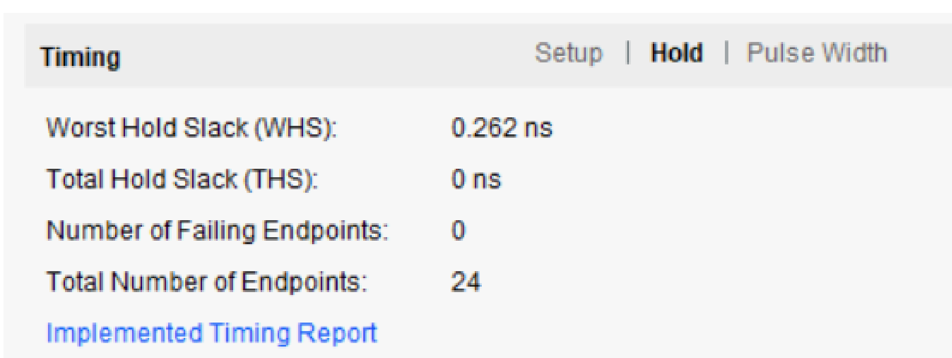

Figure 5. Hold timing summary — 0.262 ns worst hold slack, 0 failing endpoints.

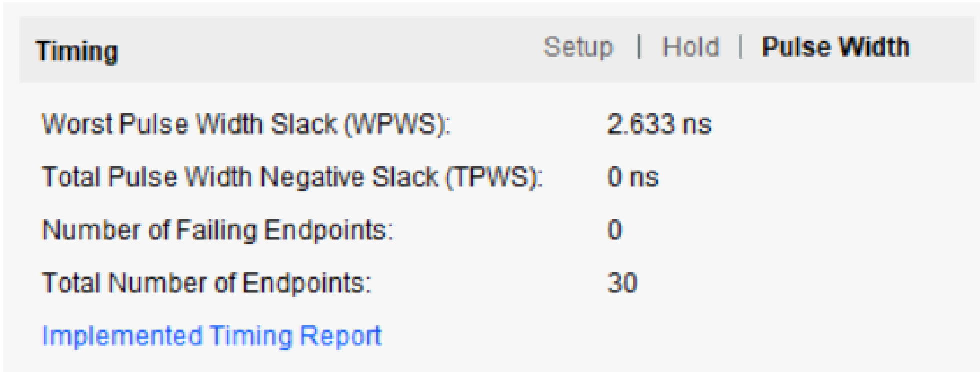

Figure 6. Pulse-width timing summary — 2.633 ns worst pulse-width slack, 0 failing endpoints (30 endpoints checked).

Check	Worst Slack	Total Negative Slack	Failing Endpoints	Total Endpoints
Setup	121.49 ns	0 ns	0	24
Hold	0.262 ns	0 ns	0	24
Pulse Width	2.633 ns	0 ns	0	30

All checks pass with zero failing endpoints and zero total negative slack, indicating a fully timing-clean implementation.

7. Functional Behavior on Hardware

With the design programmed onto the ZedBoard, the following behavior is expected and was the design intent:

- LD0 (counter[0]) toggles every 1 second.
- LD1 (counter[1]) toggles every 2 seconds.
- LD2 (counter[2]) toggles every 4 seconds.
- Together the three LEDs display the binary sequence 000 → 001 → 010 → ... → 111 → 000, incrementing once per second and completing a full 8-second cycle.
- Pressing/releasing the reset switch (SW0, active-low) asynchronously clears the divider and counter back to 000, once the PLL's locked signal is high.